

LECTURE CHALLENGE – compare a summer day to a winter day in Romania

GROUP no ……

Summer day in Romania	**Winter day in Romania**
p = 1 bar	p = 1 bar
t = 30°C	t = 0°C
φ = 60%	φ = 85%
$p_{sat}(t)=$	$p_{sat}(t)=$
$p_v = \varphi * p_{sat}(t)=$	$p_v =$
$x = 0.622 \frac{p_v}{p-p_v} =$	$x =$
$t_{dp} = t_{sat}\ (p_v)$	$t_{dp} =$
$h= t + x(2500 + 1.84t) =$	$h=$

Known data for water at saturation:

t [°C]	**-6**	**-4**	**-2**	**0**	**2**	**4**	**6**	**8**	**10**	**12**	**14**	**16**
p_{sat} **[Pa]**	390.9	454.9	527.9	611.2	706	813.5	935.3	1073	1228	1403	1599	1819
t [°C]	**18**	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**	**40**
p_{sat} **[Pa]**	2065	2339	2645	2986	3364	3783	4247	4760	5325	5948	6633	7385

Individual work to build together an entire day simulation!

Bucharest – Meteornorm data, August 8th versus February 1st

Summer day			Winter day		
Time, UTC	**t [°C]**	**φ [%]**	**Time**	**t [°C]**	**φ [%]**
1	22.5	83	1	3.5	79
2	21.8	89	2	2.9	81
3	21	90	3	2.2	84
4	21	91	4	1.9	81
5	20.7	89	5	1.6	83
6	20.9	86	6	1.3	83
7	22	80	7	1.1	85
8	23.3	80	8	1.4	84
9	24.4	62	9	3.3	77
10	25.6	55	10	5.3	72
11	26.3	54	11	6.7	73
12	26.9	53	12	7.4	74
13	27.3	48	13	7.9	69
14	27.7	47	14	7.9	66
15	27.9	46	15	7.7	67
16	27.9	44	16	7	74
17	27.3	56	17	5.8	78
18	26.7	46	18	5.3	77
19	25.9	49	19	4.8	77
20	24.6	53	20	4.3	81
21	23.3	53	21	3.8	82
22	22	55	22	3.3	87
23	20.6	59	23	2.8	86
24	19.3	60	24	2.3	85